Christos Hadjinikolis

Engineering Manager | Production ML & Real-Time Data Systems
Ph.D. Computer Science | Engineering Leadership, Streaming Systems & AI Standards
✉ christos.hadjinikolis@gmail.com

profiles
website
LinkedIn
GitHub
Google Scholar

leadership
6 direct reports
10-person team
hiring/coaching
delivery ownership

systems
real-time data
Kafka/Flink
model/API serving
observability

ML product
production ML
eval/replay
data contracts
model monitoring

maturity
ADRs/forums
runbooks/on-call
local E2E tests
rollback/fallback

domain
commodity intel
maritime data
AI standards
explainability

tooling
PyPI: dynamicio
skeleton-replay
JetBrains plugin

about me

I am an **Engineering Manager and ML systems lead** with 16 years across applied AI, data platforms, back-end services, consulting, and academic research. I lead cross-functional teams that turn research-grade models, high-volume data, and ambiguous product requirements into reliable production systems. Engineering Manager since February 2022, following industry ML engineering and consulting roles since 2016. My work combines people development, architecture, delivery ownership, streaming infrastructure, model/API serving, evaluation and operational reliability.

work-experience

Vortexa Ltd, London, UK 12/2020–present — Engineering Manager | ML Systems Lead

I lead technical direction and delivery for Vortexa's live post-ingestion ML/data estate, turning **6M** vessel-position records/hour into production commodity intelligence for **13.5K** monitored vessels. Manage **6** direct reports and lead a **10-person** cross-functional team accountable for model quality, data-product interfaces, client and stakeholder clarity, reliability, and delivery maturity. Joined as Senior ML Engineer in December 2020; Engineering Manager / ML Systems Lead since February 2022.

- **People & Delivery:** Manage **6** direct reports across Product, SME analysis, Data Science, and Data Engineering; own performance reviews, career development and delivery. Hiring manager for **6** roles; shaped onboarding and hiring practices as Data Production grew from **4 to 30+**. Retained the full team and supported every member's promotion or progression (July 2026 snapshot).
- **Technical Direction:** Shape architecture and engineering direction across model/API serving, streaming pipelines, evaluation/replay interfaces, data contracts, and downstream product integrations; serve as a senior technical escalation point while remaining close to implementation and long-term system evolution.
- **Production ML Systems:** Led research-to-production delivery for destination and arrival-time sequence/transformer models in **PyTorch**; established MLflow, model/data versioning, evaluation, replay, monitoring and model-serving workflows.
- **Streaming & Distributed Processing:** Led the **Kafka Streams-to-Flink** migration as a strategic platform move, improving operational visibility, state/checkpointing, maintainability, and compute scaling independent of Kafka partitioning.
- **Reliability & Observability:** Maintained signal-quality controls, live accuracy monitoring, runbooks, rollback/fallback paths, shared on-call, and **MTTR** under **30** min for production prediction and streaming systems.
- **Engineering Maturity:** Standardised I/O seams, schema/data validations, repo archetypes, ADRs, dev containers, local E2E tests, and DORA/Jira delivery-health signals; cut a **3h** DAG feedback loop to under **5** min.
- **Client & Stakeholder Alignment:** Addressed direct client and stakeholder concerns and ran workshops across Product, SMEs, analysts, and engineers to balance product pull with engineering trade-offs, turning model-quality disputes into measurable workstreams.

DataReply, London, UK 04/2016–12/2020 — Senior Consultant | Data Scientist / ML Engineer

Joined Data Reply's London spin-off as its first consultant, operating inside and contributing to a practice that grew from **three practitioners and two managers to 30+ consultants**. Progressed to Senior Consultant; shaped opportunities, agreed deliverables/timelines, managed expectations, placed consultants, and helped build the consulting culture around clear technical delivery across UBS, Vodafone, CNHi, and other clients.

- **Vodafone Group:** Led multi-workstream GCP data-science platform work based on Kubeflow; drove requirements, migration planning, CI/CD, workflow automation, ML lifecycle support, and feature-engineering migration.
- **CNHi:** Lead Data Scientist/Scrum Master on predictive maintenance (Azure/Databricks); translated stakeholder needs into DS/DE backlog and guided a **PySpark** team over live telemetry.
- **UBS:** Embedded consultant who grew from Data Scientist into ML Engineer through XP/pairing while delivering graph/process-mining analytics, observability, and real-time **Kafka**/Elasticsearch pipelines.

KCL, UCL, David Game College, UK 2010–2016 — Teaching and Technical Training

Delivered courses and tutorials in algorithm design, data structures, databases/SQL, and Java/Python during and after doctoral research; developed presentation discipline, learner support, and mentoring habits.

education

2010–2014 **Ph.D. in Computer Science** King's College London, UK

***Thesis**: Persuasion Dialogues & Opponent Modelling.*

Developed logical inference methods for large **Knowledge Graphs**, combining symbolic reasoning, **graph analytics**, and Bayesian techniques (e.g., MCMC). Received **Best Poster Award** at IJCAI 2013 (413 submissions), **EPSRC PhD Scholarship**, **Outstanding TA Award**, and **Graduate Certificate in Academic Practice**.

Access Publications on Google Scholar

2004–2010 **Diploma (BEng) in Computer Engineering** University of Thessaly, GR

Five-year polytechnic degree in Computer Science and Engineering with a **strong focus on mathematics** (10 courses), covering theoretical aspects, networking, and telecommunications. ***Majored in Artificial Intelligence***.

2002–2004 **Military Officer: Second Lieutenant** National Guard, CY

Completed national service as a commissioned Second Lieutenant.

selected AI/platform tooling

Promet Personal project | Local AI harness

Applied AI project exploring voice, memory, tool-use, and streaming assistant workflows; hands-on work on local runtimes, faster-whisper STT, Kokoro/**Hugging Face** assets, tool routing, schema validation, approval gates, durable state, traces, and deterministic replay/evaluation for auditable LLM interactions.

PyPI **skeleton-replay** Runtime evidence

Python developer tool that turns script/pytest runs into traces, architecture snapshots, workflow evidence, and replayable reports for review, debugging, onboarding, and evidence-led code understanding.

JetBrains **Skeleton Replay plugin** IDE workflow

Free PyCharm/IntelliJ plugin bringing Skeleton reports, workflow evidence, and source navigation into the IDE; reflects a bias for developer experience, evidence, and shared understanding.

PyPI **dynamicio** Data contracts

Free public tooling for making I/O seams explicit, validating schema/data expectations, and switching local/dev/prod datasets so pipeline logic can be tested with faster feedback.

committees/affiliations

since 10/2024 **University College London, Dept. of Information Studies** Associate Researcher

Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.

since 2021 **ISO - International Organisation for Standardisation** Committee Expert Member for JTC'21

Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.

talks/interviews

2023 **Bill Raymonds, Agile in Action** Podcast Interview

Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa, **Details:** YouTube

2022 **Open Data Science Conference** Industry Conference Talk

dynamicio, a published **PyPI library** for abstracting I/O in ML/data workflows, **Details:** here

2020 **iunera: Big Data Innovators** Interview Blog Post

The Agile approach in data science explained by an ML expert, **Details:** here

2020 **Big Data Warsaw** Industry Conference Talk

Monitoring & Analysing Communication and Trade Events as Graphs, **Details:** here

2018 **Connected Data London** Panelist @ Industry Conference

AI tribes and interconnections: what's graph got to do with it?, **Details:** here

2018 **Minds Mastering Machines** Industry Conference Talk

A talk on doing Data Science the Agile Way, **Details:** here

certificates & awards

2017–2020 **Selected ML/cloud credentials** AWS, Google Cloud, O'Reilly

AWS ML Specialty; Google Professional Data Engineer; Apache Spark Developer.

2016–2019 **Data/graph engineering credentials** Coursera, Elastic, Neo4j

Process Mining; Graph Analytics; Advanced Elasticsearch; Neo4j Certified Professional.